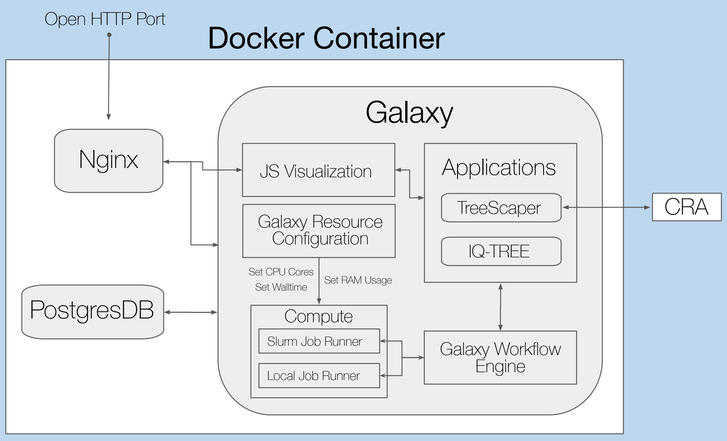
CloudForest: Dockerized Galaxy as a Platform for an Integrated Phylogenomics Software Workflow

Reid Wagner [1], Benjamin S. Toups [2], Zhifeng Deng [3], Kyle A. Gallivan [3], Jeremy M. Brown [2], James C. Wilgenbusch [1]

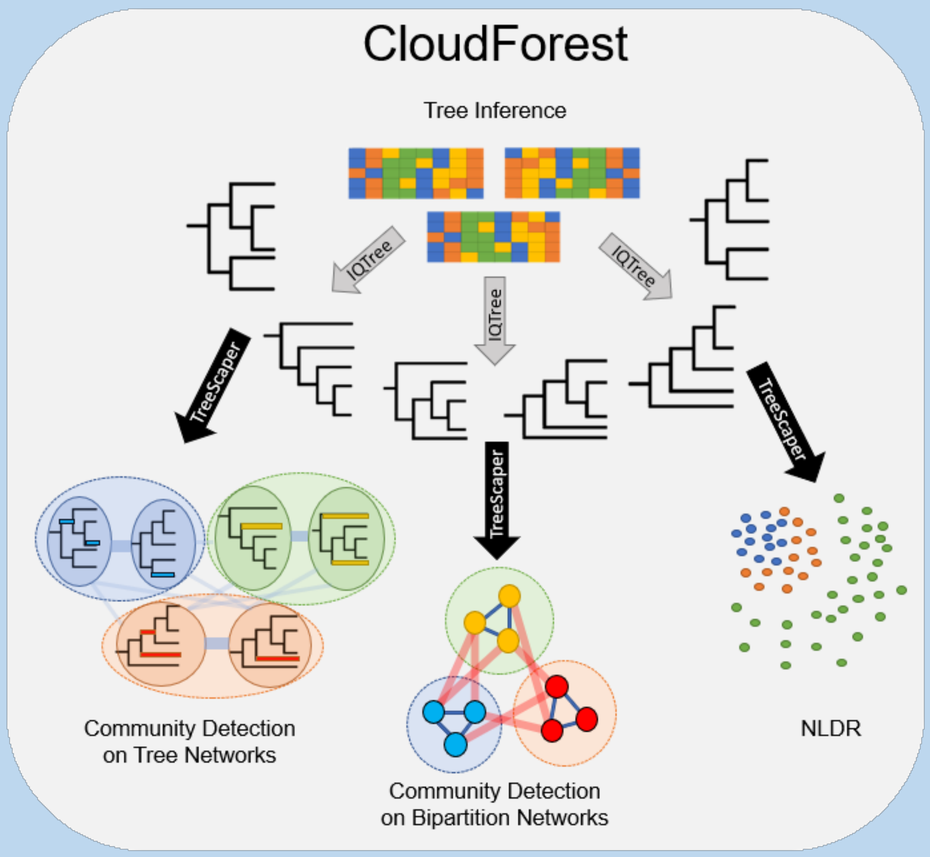
[1] University of Minnesota, [2] Louisiana State University, [3] Florida State University

CloudForest seamlessly integrates multiple tools to form a **flexible** and **dynamic** Galaxy-based platform for phylogenomic inference.

Architecture

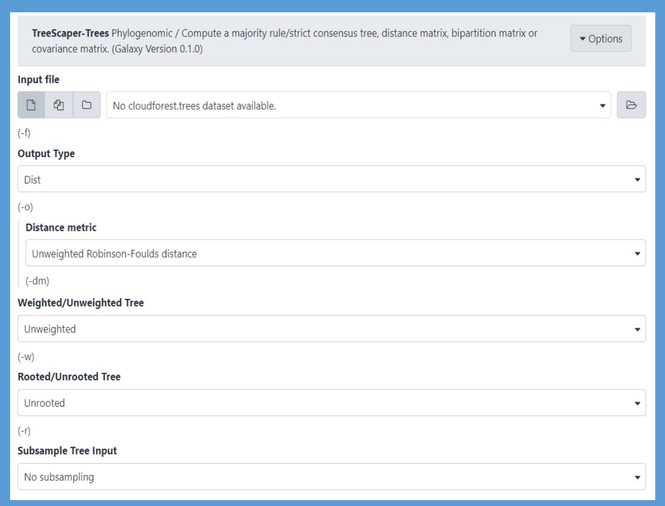


- CloudForest is centrally housed within the Galaxy Framework due largely to its large user and developer base, as well as its selection of ready-made features that lend themselves particularly well to the uses of CloudForest.
- When using CloudForest in a Docker container, local CPU and RAM resources can be allocated to the container based on the user's needs.
- Planned integration of the Slurm Workload Manager will allow users to access HPC resources via CloudForest for large scale phylogenomic analyses.
- CloudForest can offload tree inference computation to remote HPC via the CIPRES REST API (CRA) or run inference with a separate Galaxy tool. For example, the IQ-TREE Galaxy tool comes pre-installed in the CloudForest image.

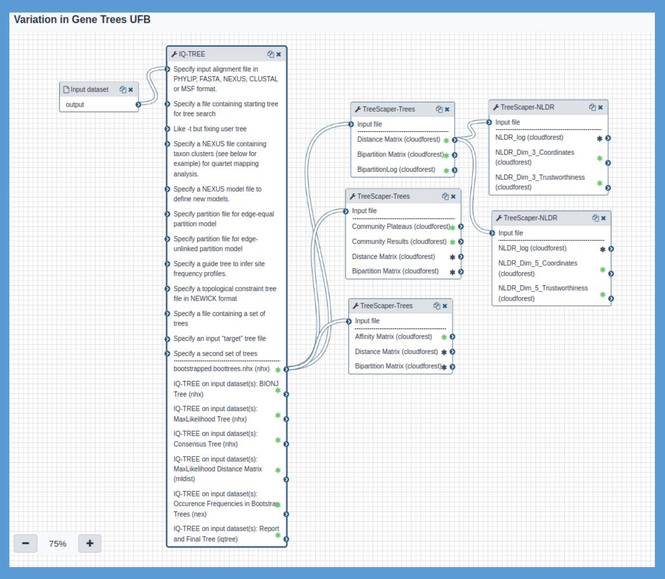


Large sets of trees often contain a vast amount of information about the dataset simply in the way that treespace itself is organized. However, the means with which to collect and analyze this information are currently disjointed and limited in their scope. CloudForest seeks to address this by integrating and streamlining multiple programs required to perform such analyses in an intuitive way, and to add functionality to address gaps in the current software.

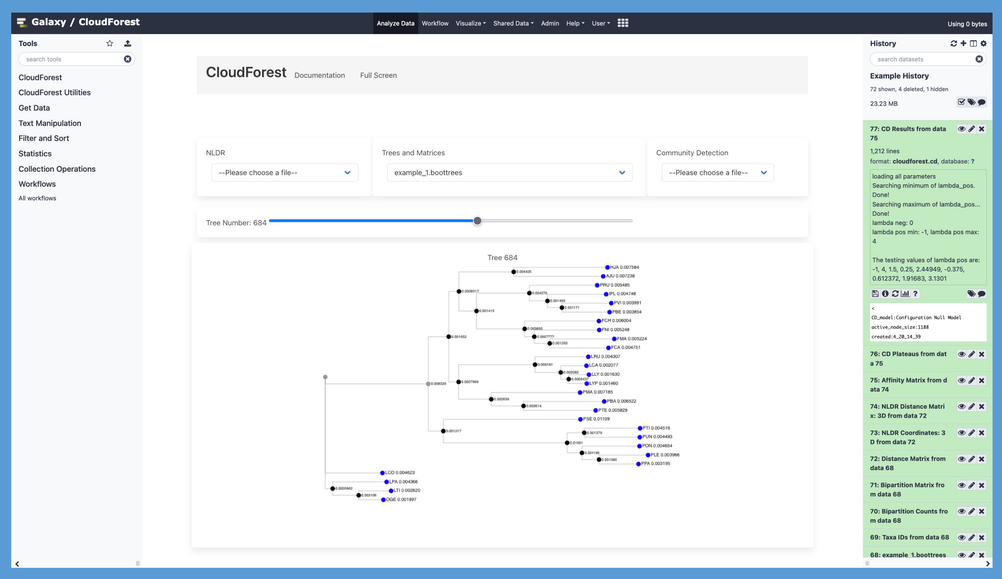
Interface



Galaxy's easy to use menus and interfaces lower the barrier to entry compared to configuring options for the underlying command-line tools.



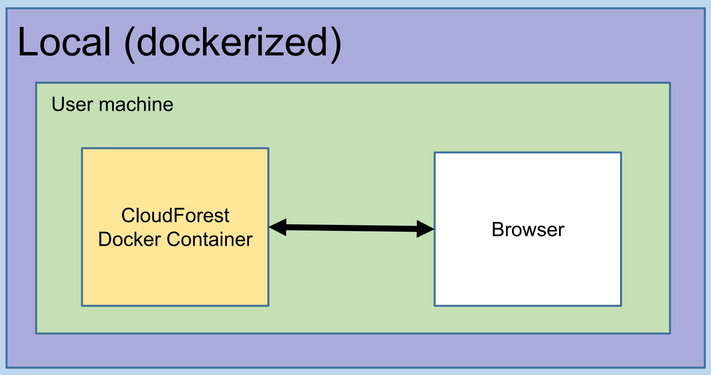
Within CloudForest, TreeScaper is split into a suite of different tool wrappers for its different features. Galaxy workflows can represent the relationships between inputs and outputs of different computations and allow reproducible multi-step analyses.

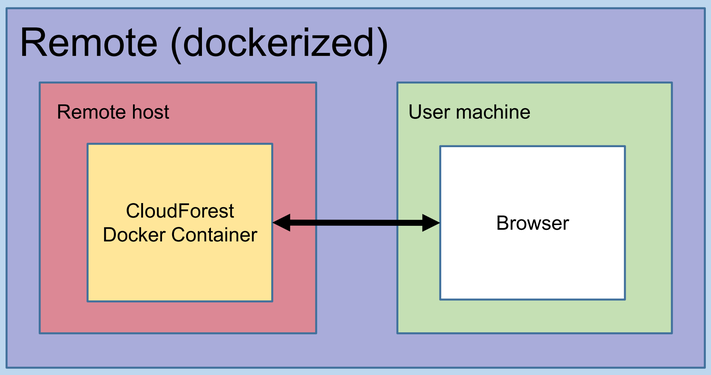


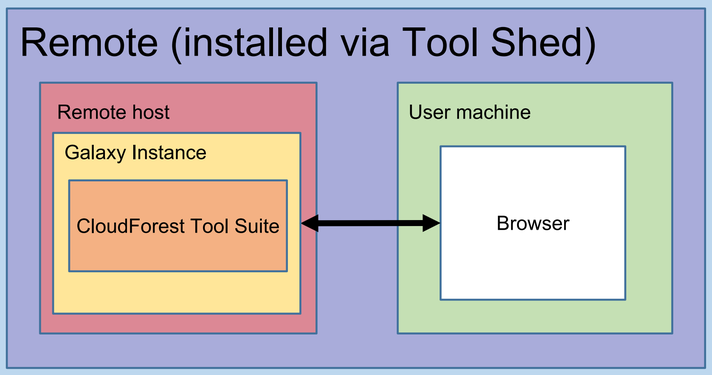
CloudForest includes phylogenomic visualizations, built with HTML5 and JavaScript libraries D3.js and Plotly, as Galaxy plugins alongside the provided Galaxy tools. This allows users to more easily examine their data in a self-contained ecosystem.

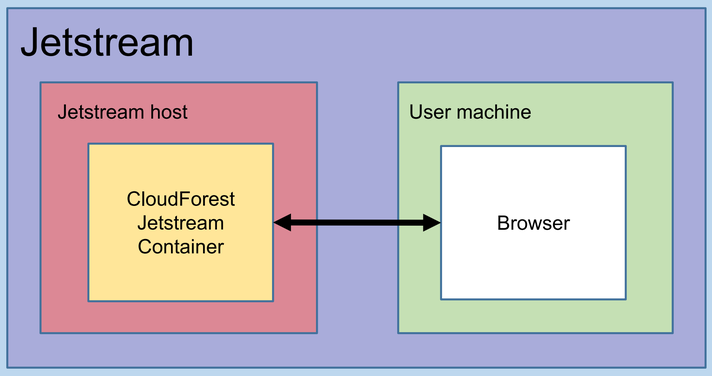
Deployment

- CloudForest is available in a variety of formats to best suit the needs of researchers. Currently, users can run CloudForest on a remote server or through a locally stored Docker image.
- Future plans include packaging the CloudForest tool suite for the Galaxy Tool Shed, as well as creating a dedicated CloudForest Jetstream image.









CloudForest is actively seeking out beta testers to use our platform with phylogenomic datasets to answer novel questions and provide feedback. Reach out to our team today if you are interested in becoming a beta tester!

References and Acknowledgements

- CloudForest is supported by NSF grants DBI-1934182 to JCW, DBI-1934157 to KAG, and DBI-1934156 to JMB.
- T. McGowan contributed significantly to the original design and implementation of CloudForest.
- M. Miller, W. Pfeiffer, and M. Zhuang provided valuable guidance in integrating the CRA with CloudForest.
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